Haryana LEET / OCET 2026

Section B — Electronics Stream Courses

25 Questions | 25 Marks | No Negative Marking

b-1 ELEMENTS OF ELECTRICAL ENGINEERING (9 Q | 9 M)

■ EMF, Kirchhoff's Laws & Network Theorems

- KCL: $\Sigma I_{in} = \Sigma I_{out}$ at a node
- KVL: $\Sigma V = 0$ around a closed loop
- Thevenin's theorem: V_{th} & R_{th} | Norton's theorem
- Superposition theorem | Max. power transfer theorem
- Faraday's laws of electromagnetic induction

■ Magnetic Circuits

- MMF, reluctance, flux — analogy with electric circuit
- Self inductance L | Mutual inductance M
- Lenz's law — direction of induced EMF

■ AC Circuits

- RMS value = Peak/$\sqrt{2}$ | Average value = 0.637 × Peak
- Form factor = RMS/Average = 1.11
- Impedance $Z = \sqrt{R^2+(X_L-X_c)^2}$ | Phase angle ϕ
- Power factor $\cos\phi = R/Z$ | Apparent, Active, Reactive power
- Series RLC: V leads I (inductive) / V lags I (capacitive)

■ Resonance

- Series resonance: $f_r = 1/(2\pi\sqrt{LC})$ | $Z_{min} = R$ at resonance
- Q-factor (quality factor) = $\omega L/R$
- Parallel resonance: high impedance at resonance

■ Transformers

- EMF equation: $E = 4.44\ f\ N\ \Phi_m$
- Turns ratio: $N_1/N_2 = V_1/V_2 = I_2/I_1$
- Efficiency $\eta = P_{out}/P_{in}$ | Losses: copper & core (iron)
- 3-phase transformer: star-delta, delta-star connections

■ DC Machines

- DC Generator: EMF = $P\Phi NZ/60A$ (lap: A=P, wave: A=2)
- Types: separately excited, shunt, series, compound
- DC Motor: back EMF $E_b = V - I_aR_a$ | $N \propto E_b/\Phi$
- Motor types & characteristics: shunt (const. speed), series (high torque)

■ 3-Phase Induction Motor

- Rotating magnetic field — synchronous speed $N_s = 120f/P$
- Slip $s = (N_s-N)/N_s \times 100\%$ | Rotor frequency = sf
- Torque ∝ sE■²R■ / (R■²+s²X■²)

■ Transmission & Distribution

- Advantage of HV: lower current → lower I^2R losses
- 3-phase vs single-phase: 3ϕ more economical, delivers more power
- 3-wire DC vs 2-wire DC comparison

b-2 ELEMENTS OF ELECTRONICS ENGINEERING (8 Q | 8 M)

■ Semiconductor Diodes

- PN junction: depletion layer, barrier potential
- Forward bias: current flows | Reverse bias: only leakage
- Zener diode: breakdown voltage, voltage regulation circuit
- LED: emits light when forward biased | Photodiode: reverse biased

■ BJT (Bipolar Junction Transistor)

- NPN & PNP | 3 regions: emitter, base, collector
- Configurations: CB, CE, CC — characteristics & comparison
- CE: highest current gain β = IC/IB | Most common
- Biasing: fixed bias, voltage divider bias — Q-point stabilization

■ FET & MOSFET

- JFET: N-channel, P-channel | Gate voltage controls drain current
- Enhancement & depletion MOSFET | MOSFET in digital circuits

■ Amplifiers & OPAMP

- Small signal amplifier: voltage gain Av = Vout/Vin
- OPAMP characteristics: very high Av, high Rin, low Rout
- Inverting amplifier: Av = -Rf/Rin | Non-inverting: Av = 1+Rf/Rin
- Summing amplifier, integrator, differentiator circuits
- Oscillators: Barkhausen criterion ($A\beta$=1, phase=0°)

■ Digital Electronics — Most Important

- Number systems: Binary, Octal, Hexadecimal ↔ Decimal conversions
- Binary arithmetic: addition, subtraction (2's complement)
- BCD, Gray code, excess-3 code conversions
- Logic gates: AND/OR/NOT/NAND/NOR/XOR/XNOR — truth tables
- Boolean algebra: laws, De Morgan's theorem
- Simplification using K-map (2,3,4 variable)
- Flip-Flops: SR, JK, D, T — truth table, clocking

■ Communication Systems

- Modulation need: low freq signal cannot travel long distance
- AM: carrier amplitude varies | FM: carrier frequency varies | PM: phase varies
- Bandwidth of AM = 2fm | Modulation index m = Am/Ac
- Demodulation: envelope detector (AM), discriminator (FM)

■ Measurements & Instrumentation

- CRO: measure voltage, current, frequency, phase — operation
- Ammeter (series) vs Voltmeter (parallel) connection
- Wattmeter, energy meter (kWh) working principle
- Multimeter: measure V, I, R | Insulation tester, Earth tester
- Errors: gross, systematic, random | Accuracy vs precision
- Transducers: thermistor (temp), LVDT (displacement)

■ Power Electronics

- SCR (Silicon Controlled Rectifier): 4-layer PNPN device
- SCR triggering: gate pulse | Applications: speed control, converters
- DIAC: bidirectional — triggers at breakover voltage
- TRIAC: bidirectional SCR — used in AC power control

b-3 ELEMENTS OF COMPUTER ENGINEERING (8 Q | 8 M)

■ Computer Fundamentals

- Von Neumann architecture: CPU, Memory, I/O
- ALU, CU, Registers — functions
- Compiler: translates entire HLL program to machine code
- Interpreter: translates & executes line by line
- Assembler: converts assembly language to machine code
- Algorithm properties: Input, Output, Definiteness, Finiteness, Effectiveness

■ Computer Organization

- Registers: PC, IR, MAR, MDR, AC, SP
- Bus: data bus, address bus, control bus
- Instruction cycle: Fetch → Decode → Execute
- Hardwired vs Micro-programmed control unit
- Cache memory: L1, L2, L3 | Virtual memory: paging, segmentation

■ Operating Systems

- Functions: process management, memory management, I/O
- Process states: new, ready, running, waiting, terminated
- CPU scheduling: FCFS, SJF, Round Robin, Priority
- Critical section problem | Semaphore, Mutex
- Deadlock: conditions (mutual exclusion, hold & wait, no preemption, circular wait)

■ C Programming — Most Asked

- Data types: int, float, double, char, long | sizeof() operator
- Operators: arithmetic, relational, logical, bitwise, assignment, ternary
- Control statements: if, if-else, nested if, switch-case
- Loops: for, while, do-while | break, continue, goto
- Arrays: 1D, 2D — declaration, initialization, traversal
- Strings: strlen, strcpy, strcat, strcmp functions
- Pointers: declaration, dereferencing (*p), pointer arithmetic
- Functions: call by value vs call by reference | Recursion
- Structures & Unions: definition, accessing members (. and ->)
- File handling: fopen, fclose, fprintf, fscanf, fread, fwrite

■ Networking Basics

- LAN: limited area, high speed | WAN: wide area, uses routers
- MAN: metropolitan area | Internet: global WAN
- TCP/IP model: 4 layers | OSI model: 7 layers (names)
- IP address: IPv4 (32-bit) | MAC address (48-bit)
- HTTP, FTP, SMTP, DNS — protocol functions